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Divergent regional trends in alpine tundra productivity linked to changes in snow-free season length and summer warming

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Abstract

Despite widespread reports of tundra ‘greening’, it remains unclear whether greening is occurring in mid-latitude alpine tundra of the Southwestern United States, where vegetation productivity strongly influences headwater ecohydrology and ecosystem function. Using 39 years of Landsat data (1984–2023), experimental manipulations, and field measurements at the Niwot Ridge Long-Term Ecological Research site in Colorado, USA, we investigated how climate change is reshaping alpine greenness across the Southern Rocky Mountain alpine zone. Contrary to widespread greening trends documented in Arctic and European alpine tundra systems, normalized difference vegetation index (NDVI) declined 2.8% across this region, on average. Within the region, we found evidence of pronounced spatial heterogeneity in NDVI shifts, with 20.2% of the region browning while 17.6% greened. A four-year factorial snow and temperature manipulation experiment found that warming reduces NDVI while snow addition has a slight positive effect on NDVI, suggesting that soil moisture limitation can constrain productivity gains under warming in this semi-arid system. Regionally, summer warming increased the probability of either greening or browning, and the length of the snow-free season determined the direction of this relationship: browning was associated with lengthening growing seasons while greening occurred where growing seasons remained stable. These findings demonstrate that alpine productivity responses to climate change are not globally uniform and that regional context mediates temperature effects.

1. Introduction

Climate change poses a major threat to the physical, hydrological, and biological structure of Arctic and alpine tundra ecosystems. Rising spring and summer temperatures modulate snowmelt, changing the timing and amount of soil moisture during the growing season (Jay *et al* 2023, Rawlins and Karmalkar 2024). As a result, both warming temperatures and changing snowpack can alter vegetation dynamics and ecohydrological processes (Wipf *et al* 2009, Migliavacca *et al* 2012, Kelsey *et al* 2021, Wieder *et al* 2022, Choler *et al* 2024). One salient example of these impacts is the phenomenon of ‘tundra greening’—increases in normalized difference vegetation index (NDVI) or other remotely-sensed vegetation indices—that has been detected across large swaths of the Arctic and Eurasian alpine tundra over recent decades (Zhu *et al* 2016, Rumpf *et al* 2022, Choler *et al* 2024, Frost *et al* 2025).

Analyses using satellite data have generally detected more greening (apparent increasing productivity) than browning (apparent decreasing productivity) in both Arctic and alpine tundra regions, but the patterns are heterogeneous across space and time (Myers-Smith *et al* 2020). A recent pan-Arctic study (Berner *et al* 2020) determined that approximately 37% of the Arctic tundra experienced greening between 1985 and 2016—primarily associated with increased summer temperatures. Browning was

documented in only 5% of the region and the remaining area (58%) exhibited no detectable change in greenness over this period. Regional studies in select areas suggest similar processes are likely occurring in alpine tundra. For example, 70%–80% of the Eurasian alpine tundra has greened over recent decades whereas <1% has browned, a pattern seemingly linked to earlier snowmelt (Rumpf *et al* 2022, Choler *et al* 2024). However, greening trends in High Mountain Asia (i.e. the Tibetan Plateau, Tianshans, Pamir Plateau, Kunlun Mountains and Qilian Mountains) that were prevalent between 1980 and 2000 have shown slight browning in recent decades, likely due to moisture stress accompanying higher rates of evapotranspiration (Liu *et al* 2021). Thus, while warming can increase productivity both directly (through increased plant growth) and indirectly (through warming-induced earlier snowmelt, which can extend the duration of the growing season), micro- and regional climates can lead to divergent productivity responses to longer and warmer growing seasons when other resources, such as soil moisture, become limiting (Ganjurjav 2016, Winkler *et al* 2016, Ma *et al* 2017, Jabis *et al* 2020, Kelsey *et al* 2021). Consequently, while greening is the dominant response to climate change that has been documented in the two most well-studied tundra regions, the Arctic and European Alps, the combined mechanistic drivers of temperature and moisture suggest that alpine tundra regions worldwide could be experiencing browning, greening or both.

Arctic and alpine tundra share similar vegetation and floristic associations, low growing-season temperatures, and slow rates of primary production and nutrient cycling (Bowman and Seastedt 2001). However, substantial differences in their physical environments mean that findings from one system may not translate directly to the other. Relative to the Arctic, mid-latitude alpine ecosystems experience higher solar irradiation, shorter photoperiods, greater vapor pressure deficits, stronger winds, less permafrost, and consequently drier soils (Billings 1973, Bowman and Seastedt 2001). These differences are particularly consequential for productivity responses to warming and altered snowpack and snowmelt. Where Arctic greening has been linked primarily to warming-driven shrub expansion facilitated by relatively moist, permafrost-supported soils (Frost *et al* 2025), alpine tundra in temperate regions such as the Southern Rocky Mountains may respond quite differently, with seasonal water availability potentially constraining the productivity gains that a longer snow-free season would otherwise promote.

The Southern Rocky Mountains (interchangeably, ‘Southern Rockies’; figure 1) form the headwaters of major rivers, including the Colorado River, a crucial water source for over 40 million people across the United States and Mexico and a basin home to many federally recognized Indigenous tribes (Udall and Overpeck 2017). Reductions in snow depth, persistence, and cover have been reported in the southernmost ranges, such as the Upper Rio Grande Basin headwaters, including the San Juan and Sangre de Cristo mountain ranges (Elias *et al* 2021). Recent observations have documented substantial declines in summer soil moisture in the Southern Rockies, and simulations of future climates predict these trends to continue throughout this century (Dettinger *et al* 2015, Bolinger *et al* 2023). Feedbacks between declining soil moisture and vegetation could exacerbate or ameliorate changes to the hydrologic cycle and surface energy balance. For example, snow cover is an important control of the growing season for alpine vegetation, with earlier snowmelt potentially providing more time for alpine plant vegetation to grow. However, taller tundra plants modify the distribution of snow and surface-atmosphere radiative exchanges, which can cause either faster snowmelt (through decreased albedo) or delayed snow-off (through a snow trapping effect; Pearson *et al* (2013), Loranty *et al* (2018)).

Despite the social and economic importance of the Southern Rocky Mountain alpine ecosystem, investigations into climate-change induced changes in productivity in this region have been limited and small in scale. An infrared heating experiment found that vegetation phenology advanced with earlier snowmelt, but soil moisture limitations reduced community productivity (Winkler *et al* 2016, Jabis *et al* 2020). Using Landsat data collected over an intensively studied 15 km² area in southwestern Colorado, Wainwright *et al* (2020) found that earlier snowmelt and reductions in soil moisture early in the growing season limited peak productivity. Together, the studies suggest that the relatively dry conditions in the Southern Rockies may generate a different response to early snowmelt and summer warming than many of the Arctic and alpine tundra regions that have been studied to date. Thus, an analysis of the patterns of greening and browning, and associated trends in summer warmth and snowmelt timing, in the Southern Rocky Mountain regions provides an important test of the generality of greening trends in other alpine regions while further examining the role of snow cover in mediating the effects of warming on alpine plant productivity.

Here, we use a combination of long-term experiments, long-term monitoring, and remote sensing datasets to evaluate trends in alpine tundra greenness in the Southern Rocky Mountains and their relationships with warming and the length of the snow-free season. Specifically, we integrate data from a four-year, factorial snow and temperature manipulation experiment at the Niwot Ridge Long Term Ecological Research site (Niwot Ridge LTER, Colorado, USA) with 20 years of point-scale alpine biomass

field measurements, 39 years of Landsat data, and historical climate records to address three questions about the causes of productivity variation in the Southern Rocky Mountain region:

- Q1. How do experimental manipulations of spring snow-depth and growing-season temperature alter alpine productivity in a focal site in the Southern Rocky Mountains?
- Q2. Are the Southern Rocky Mountains greening or browning?
- Q3. Do changes in snow cover duration and summer temperature predict spatial variation in greenness trends in the Southern Rockies?

2. Materials and methods

2.1. Study domain

2.1.1. Southern Rocky Mountains

The Southern Rocky Mountains are the southernmost segment of the Rocky Mountain system within the North American Cordillera. The North-South oriented province extends 1000 km from the southernmost reaches of the Sangre de Cristo Mountains near Sante Fe, New Mexico (35°N) north into the Laramie Mountains of Wyoming (42°N), spanning Colorado, New Mexico, Wyoming, and Utah (Coblentz *et al* 2011). In contrast to systems like the Alps (and many other high mountain systems), the Southern Rockies have relatively flat and convex ridges with subjacent steep, but not vertical, relief into glaciated valleys that lead up to the nival zone on higher and comparatively narrow, rocky peaks. This terrain, described as ‘Rocky Mountain Relief’ (Barsch and Caine 1984), forms the North American Continental Divide. The alpine zone of the Southern Rockies occupies nearly 10 000 km² of land area between ≈3300 m and 4400 m in elevation, and includes all of the alpine in Colorado and patches in New Mexico, Utah, and Wyoming (figure 1). This ecosystem is characterized by relatively extreme climatic conditions, including long, cold winters and short, cool summers. Complex topography generates ecological heterogeneity at local scales in the alpine zone, where exposed, dry, windswept tundra can be within meters of low-lying, moist or wet meadow tundra with high groundwater tables or nearby melting snowbeds that keep soils saturated for much of the growing season.

Although permafrost is likely present in patchy locations throughout the region, its extent and distribution remain uncertain. Mid-20th century observational studies reported frozen ground at depths of 1–4 m in portions of the Front Range (Ives and Fahey 1971), yet more recent field measurements have failed to detect permafrost at depths of up to 10 m in areas where models predict it should occur (Janke 2005, Leopold *et al* 2010, Janke *et al* 2012). Where permafrost does persist, it is likely deep, discontinuous, and well below the plant rooting zones in the Southern Rocky Mountains.

2.1.2. Niwot ridge long-term ecological research site

In situ data were collected from the ‘Saddle’ area in the alpine zone at the Niwot Ridge LTER site (figure 1(b)), about 60 km northwest of Denver, Colorado, USA (40°3′21.2862″N 105°35′21.1086″W, 3528 m.s.l.). The Saddle catchment consists of hilly terrain and relict sorted polygons (periglacial structures formed by past freeze-thaw-driven particle sorting) that create a patterned and vegetated land surface (Caine and Bowman and Seastedt 2001), supporting plant communities that range from dry fellfield to wet meadow and shrub tundra. This site has a relatively short growing season, typically 2–3 months in length, and considerable topographic heterogeneity that modulates wind exposure and solar irradiance, which in turn causes spatial variation in snow accumulation and the timing of snowmelt. As a result, patterns of soil moisture, snow persistence, and growing season length vary over micro topographical (sub 30 m) scales (Bowman and Seastedt 2001, Hoover *et al* 2014, Knowles *et al* 2015, Bueno de Mesquita *et al* 2018, Badger *et al* 2021).

We used data from both regional (Southern Rocky Mountain) and local (Niwot Ridge) study areas to characterize the trends and drivers of above-ground net primary productivity in the alpine (table 1). An overview of the workflow can be found in figure 2.

2.2. Quantifying alpine productivity responses to experimental snow and temperature manipulation

To examine the effects of experimental manipulations of snow and temperature on productivity (Q1), we analyzed 4 years of plot-scale NDVI measurements taken *in situ* with a handheld instrument in a warming and snow addition field experiment on Niwot Ridge (Suding *et al* 2024). Following the International Tundra Experiment (a.k.a. ITEX) protocol, this experiment uses open-top chambers (OTCs) to passively warm treatment plots 1–3 °C (Henry and Molau 1997, Marion *et al* 1997, Hollister 2023). At Niwot

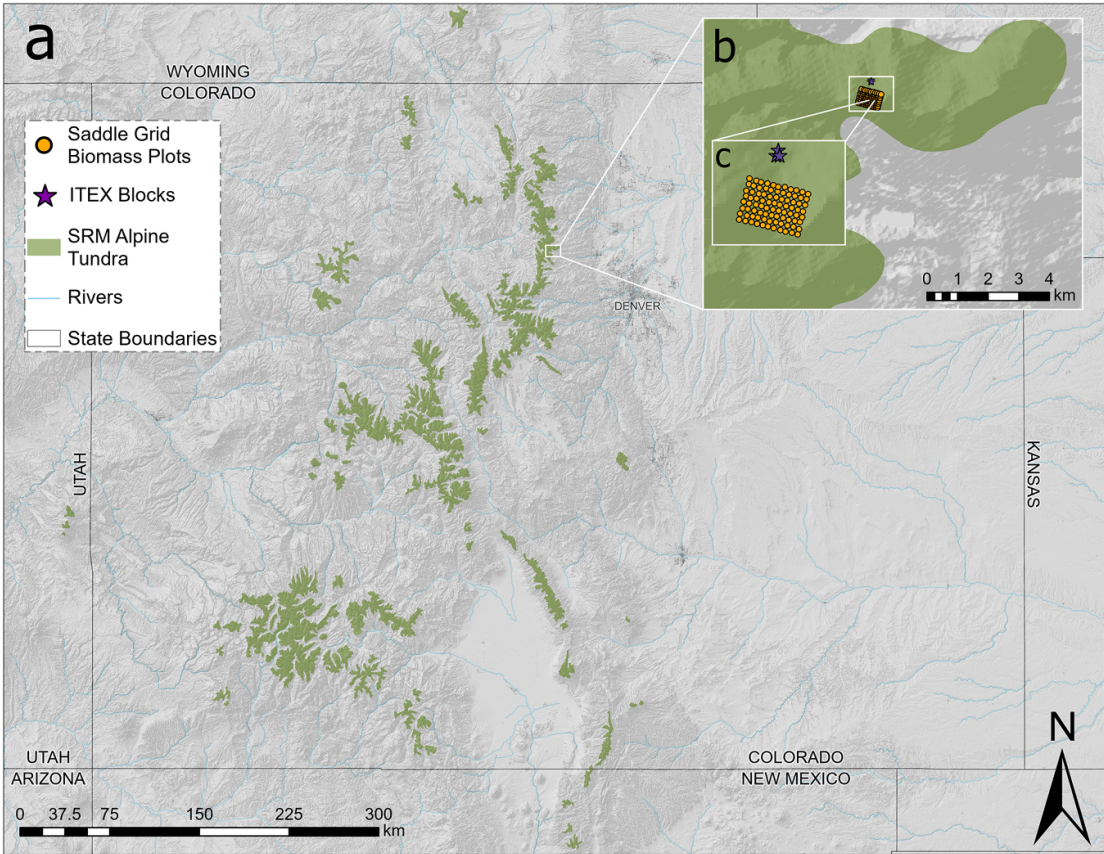


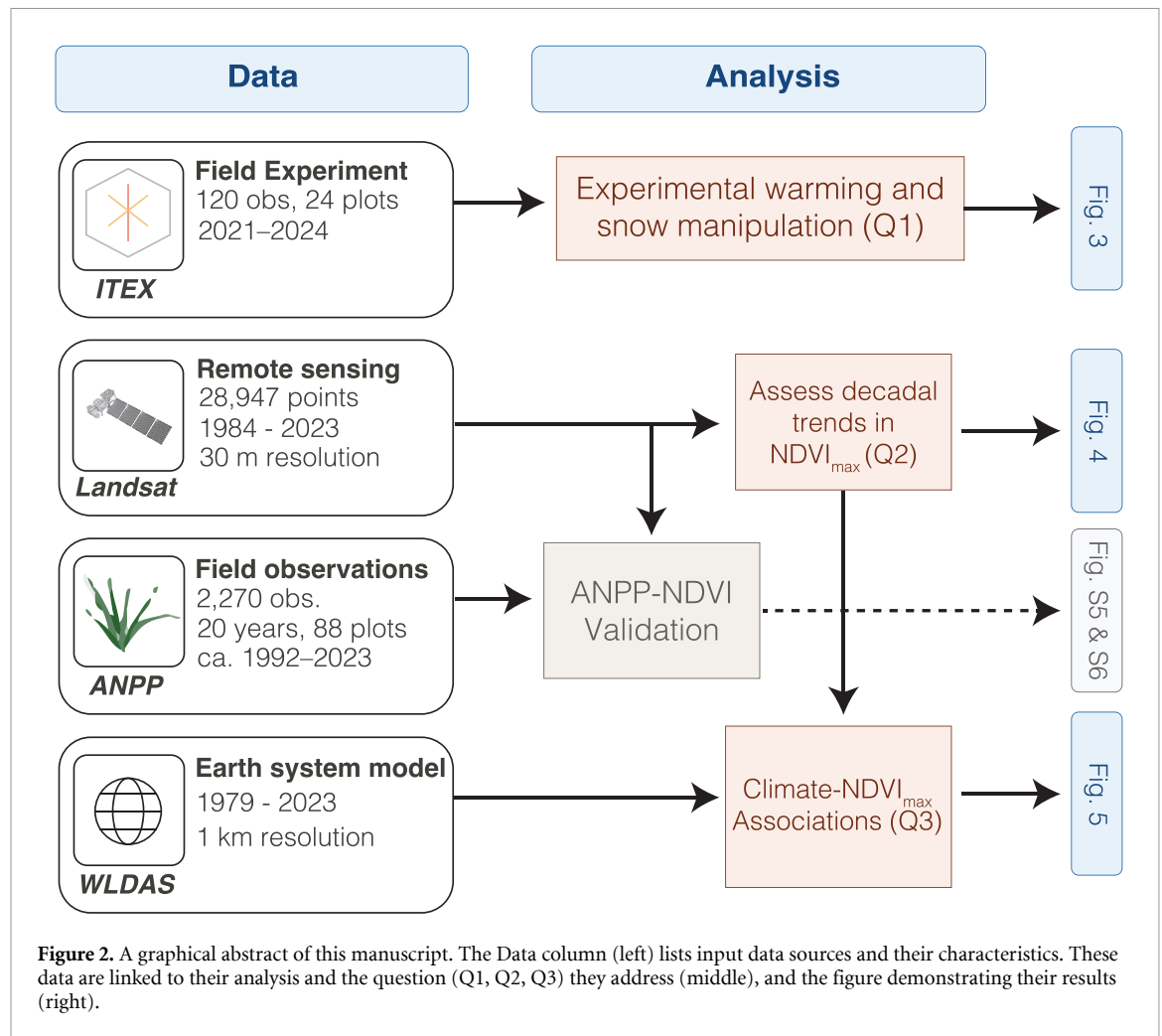
Figure 1. A map of the Southern Rocky Mountain alpine tundra. The Southern Rocky Mountain alpine zone (green shaded regions) delineated by the United States’ EPA Level IV Ecoregion classification (a). The Niwot Ridge LTER is situated in the Southern Rocky Mountain alpine with the 88 ‘Saddle grid’ long term biomass plots shown as orange circles and the ITEX blocks shown as purple stars (b), (c). Data: U.S. Census Bureau, National Elevation Dataset (10 m), U.S. EPA Ecoregions Level IV dataset (region ‘21a’), National Land Cover Database, National Weather Service.

Table 1. Summary of datasets used in this study.

Dataset	Variable	Scale	Years	References
Niwot Ridge ITEX Field Experiment	Trimble GreenSeeker NDVI	Field experiment	2021–2024	Suding <i>et al</i> (2024)
Landsat NDVI	Remotely-sensed NDVI (30 m)	Southern Rocky Mountains	1984–2023	U.S. Geological Survey
Niwot Ridge Saddle Grid ANPP	Above-ground Net Primary Productivity	Field observations	1992–2023	Walker <i>et al</i> (2023)
WLDAS Model Output	Summer temperature, Snow-cover (1 km)	Southern Rocky Mountains	1979–2023	Erlingis <i>et al</i> (2021)

Ridge, the plots are arrayed in three blocks, each containing a snow-fence that increases snow depth (leading to delayed snowmelt) in half of the plots. This factorial design contains 8 plots in each of the three blocks for a total of 24 plots with two plots per block in each treatment (Control, Snow addition only, Snow addition + OTC warming, OTC warming only). NDVI was measured with a Trimble GreenSeeker (Westminster, Colorado) handheld crop sensor once per year, approximately at peak biomass (July) each year from 2021 through 2024, except in 2021 where NDVI observations were collected in both July and August at all plots. The final dataset contains 120 plot-scale NDVI observations across all 24 plots.

Inherently, NDVI (unitless) is constrained to the closed interval $[-1,1]$. However, in practice, snow-free terrestrial surfaces are expected to have NDVI strictly in the open interval $(0,1)$, thus we fit a multi-level model with a Beta distribution and logit link function within a Bayesian framework. We model the



effect of experimental treatment conditions on NDVI, accounting for blocks, plots nested within blocks, and, separately, year as varying intercepts, resulting in equation (1):

$$\begin{aligned}
 NDVI_i &\sim \text{Beta}(\mu_i, \phi), \\
 \text{logit}(\mu_i) &= \beta_0 + \beta_1 \text{warm}_i + \beta_2 \text{snow}_i + \beta_3 (\text{warm}_i \times \text{snow}_i) \\
 &\quad + u_{b(i)}^{(\text{block})} + u_{b(i):p(i)}^{(\text{plot})} + u_{y(i)}^{(\text{year})}, \\
 \beta_k &\sim \mathcal{N}(0, 1), \quad k = 0, \dots, 3, \\
 u_j^{\text{block}} &\sim \mathcal{N}(0, \sigma_{\text{block}}^2), \quad j = 1, \dots, 3, \\
 u_j^{\text{plot}} &\sim \mathcal{N}(0, \sigma_{\text{plot}}^2), \quad j = 1, \dots, 24, \\
 u_j^{\text{year}} &\sim \mathcal{N}(0, \sigma_{\text{year}}^2), \quad j = 1, \dots, 4, \\
 \phi &\sim \text{Gamma}(4, 0.1), \\
 \sigma. &\sim \text{Student_t}^+(3, 0, 2.5)
 \end{aligned} \tag{1}$$

where $NDVI_i$ denotes the i th NDVI observation, and $\text{warm}_i, \text{snow}_i$ indicate the treatment status of observation i (these treatments are constant within plots). The terms $u_{b(i)}^{\text{block}}, u_{b(i):p(i)}^{\text{plot}}$, and $u_{y(i)}^{\text{year}}$ are varying intercepts associated with the block, the plot nested within block, and the year corresponding to observation i . Here, $b(i)$, $p(i)$, and $y(i)$ map each observation to its block, plot, and year identifiers, respectively. Priors for all models were selected based on prior predictive simulations to regularize estimates and avoid placing excessive prior probability near the boundaries of NDVI (0 and 1). Diagnostic

plots for the sampling algorithm, as well as prior and posterior predictive checks, are provided in the Supplementary Information (figures S1–S3).

To estimate the average effect of experimental manipulations while accounting for random effects, we compute empirical treatment effects (ΔNDVI) for each i th observation and modeled marginal effects $\mathbb{E}[\Delta\text{NDVI} \mid \eta_k, \eta_c]$ for each treatment k , where $k \in \{\text{snow}, \text{snow+warm}, \text{warm}\}$, using the definitions in equation (2).

$$\begin{aligned}\Delta\text{NDVI}_i &= \text{NDVI}_i - \mathbb{E}[\text{NDVI} \mid \eta_c], \\ \mathbb{E}[\Delta\text{NDVI} \mid \eta_k, \eta_c] &= \mathbb{E}[\text{NDVI} \mid \eta_k] - \mathbb{E}[\text{NDVI} \mid \eta_c].\end{aligned}\quad (2)$$

First, we calculate the expected NDVI for each experimental condition (including the control), marginalizing over the random effects. To compute empirical treatment effects (ΔNDVI_i), we subtract the expected NDVI in control plots of the same block, plot, and year ($\mathbb{E}[\text{NDVI} \mid \eta_c]$) from each observed value in treatment plots. Finally, to compute the modeled marginal effect of each treatment ($\mathbb{E}[\Delta\text{NDVI} \mid \eta_k, \eta_c]$), we subtract the expected NDVI in control plots ($\mathbb{E}[\text{NDVI} \mid \eta_c]$) from the expected NDVI value for each treatment plot ($\mathbb{E}[\text{NDVI} \mid \eta_k]$), where η_c and η_k are model parameters marginalized over random effects for control and treatment k plots, respectively.

2.3. Quantifying trends in remotely sensed greenness across the Southern Rockies

2.3.1. Landsat data

We quantified trends in annual maximum greenness values ($\text{NDVI}_{\max}$) across the entire Southern Rockies from 1984 to 2023 (Q2). To this end, we constructed a time series of Landsat spectral reflectance data. We first extracted all available Landsat 5 (Thematic Mapper, TM), 7 (Enhanced Thematic Mapper+, ETM+), and 8 (Operational Land Imager, OLI) data from 1984 to 2023 at 30 000 random points across the Southern Rocky Mountain alpine tundra (EPA Level 4 Ecoregion ‘21a’) using the LandsatTS package in R (Omernik and Griffith 2014, Gorelick *et al* 2017, Earth Resources Observation and Science (EROS) Center 2020a, 2020b, 2020c, Berner *et al* 2023). For sets of random points that were within the width of a Landsat pixel (30 m) of each other, only the first point was retained. We then filtered observations down to the growing season of each year, generously defined as days between the 152nd and 274th days of each year (approximately June through September). We removed pixels that were likely over water or snow and dropped any observations that were contaminated by more than 80% combined cloud and cloud shadow coverage using the CFMask Algorithm (Zhu and Woodcock 2012, Zhu *et al* 2015). To further minimize measurement error caused by poor lighting conditions or geospatial uncertainty, we also excluded observations acquired when the solar zenith angle was $>60^\circ$ or associated with a spatial geometric mean squared error >30 m.

The final curated dataset consisted of 5.2 million observations collected from 28 947 unique sampling locations across the Southern Rocky Mountain alpine tundra. Drawing from previous research that examined nearly 50 000 sites across the entire global Arctic biome and noting statistical stabilization in our own analysis after approximately 10 000 sites in our region (figure S7), we determined that our sample size of 28 947 sites balanced analytical precision and computational efficiency.

2.3.2. Cross-calibration of satellite data

Systematic differences in reflectance data exist among Landsat sensors because of changes in the technology with each sequential satellite. These discrepancies can introduce biases into time series that concatenate data from multiple sensors as changes in reflectance over time are confounded with radiometric differences among satellites (Sulla-Menashe *et al* 2016, Sousa and Small 2017, Berner *et al* 2020, Bayle *et al* 2024).

Landsat scenes in arctic and alpine regions are often obscured by clouds, reducing the frequency of viable scene overlaps for cross-calibrating point-vector time series. To address this issue, we used LandsatTS (Berner *et al* 2023) to cross-calibrate data and build Landsat 7 equivalent time series. LandsatTS first computes 15-day rolling mean values within satellites before fitting calibration models. We used an iterative polynomial regression approach, implemented in LandsatTS, to fit first-, second-, and third-degree polynomials and identify the best-fitting cross-calibration model with Bayesian Information Criteria. This process yielded two equations for each vegetation index that predict Landsat 7 data from Landsat 5 and 8 measurements (figure S8, and equation S4). The resulting cross-calibrated, multi-satellite time series had markedly reduced inter-sensor bias, indicating successful cross-calibration.

2.3.3. Estimating $\text{NDVI}_{\max}$ from Landsat timeseries

The high spatial resolution (30 m) and long temporal domain (1984 - present) make Landsat an attractive product for longitudinal studies in the heterogeneous terrain of the alpine tundra. However,

the temporal resolution (8- to 16-d intervals) requires additional processing to generate an unbiased annual estimate of peak NDVI. We used the LandsatTS package to estimate the annual NDVI_{max} from our Landsat time series at each of the 28 947 sites for each year between 1984 and 2023. LandsatTS accounts for both the seasonality of NDVI and biases created by an increase in observation frequency after 1999 using weighted smooth splines fit to rolling multi-year windows (Bayle *et al* 2024). We refer readers to Berner *et al* (2020) and Berner *et al* (2023) for a comprehensive description of the model algorithm.

2.3.4. Validating modeled estimates of NDVI_{max}

Field-based measurements are critical to any remote-sensing study yet are often omitted. Long-term field measurements of above-ground net primary productivity (ANPP) provide an opportunity to evaluate the extent to which satellite-derived estimates of NDVI_{max} are an effective surrogate for above-ground biomass. To guide our selection of a best-performing remote sensing metric for our study region, and ultimately justify our interpretation of changes in NDVI_{max} over time as reflecting changes in alpine vegetation, we leveraged 20 years of field-harvested ANPP data from the 88 plots distributed across an 0.5 km^2 area in the alpine tundra at the Niwot Ridge LTER, collected between 1992 and 2023 (Walker *et al* 2023). We used these data to compare four spectral vegetation indices (i.e. NDVI, EVI, EVI2, SAVI) and two methods (the LandsatTS algorithm, as described above, and a simpler method based on the 75th quantile) in modeling the spatial and temporal variation in field-harvested ANPP. We refer readers to the Supplementary Information (section SI2) for a detailed description of these analyses.

2.3.5. Identifying trends in Landsat NDVI_{max}

Data limitations occasionally restricted our ability to calculate a valid NDVI_{max} estimate in all sites in all years (see Supplemental Information for details). To assess the direction and significance of NDVI_{max} time series, we first further restricted the 28 947 unique sites to include only those whose time series contains a valid NDVI_{max} estimate for at least 80% of the years [1984–2023], and whose first and last years of NDVI_{max} estimates are within 2 years of 1984 and 2023, respectively. This final dataset contained 23 632 unique sites. Then, we use a non-parametric Mann–Kendall test for monotonic trends on pre-whitened time series and compute Sen’s slope of NDVI_{max} as implemented in the *zyp* library in R (Zhang *et al* 2000). Prewhitening time series corrects the rate of Type-I errors in temporally autocorrelated series but can reduce the statistical power of Mann–Kendall tests when a trend does exist, thus we assess significance at the $\alpha = 0.1$ level (Bayazit and Önöz 2007, Razavi and Vogel 2018, Coen *et al* 2020). We classified pixels with a positive slope in annual NDVI_{max} over time as ‘greening’, negative slopes as ‘browning’, and all other pixels as ‘no trend’. We calculated the total proportion of sites that are greening and browning ($\alpha = 0.1$) and assess 95% confidence intervals around these estimates via bootstrapping ($n = 10\,000$).

2.4. Assessing the covariation of trends in NDVI_{max} with snow-cover and summer temperature

Finally, we sought to investigate the relationship between the occurrence of greening and browning and trends in snow-cover duration and summer temperature (Q3). We extracted all available snow-cover fraction and surface temperature data from NASA’s Western Land Data Assimilation System (WLDAS). WLDAS is an instance of NASA’s Land Information System that combines land surface information, meteorological forcing data, and satellite products to model daily estimates of water and energy fluxes and reservoirs for the Western United States in a $1 \text{ km} \times 1 \text{ km}$ grid covering 1979–2023 (Erlingis *et al* 2021). WLDAS was designed for the purpose of monitoring water and energy in the Southern Rocky Mountains, Colorado River regions, and Western United States. This purpose driven design and high spatiotemporal resolution make it an ideal dataset for examining the relationship between observed NDVI trends and historical temperature and snow cover in the complex terrain of the Southern Rockies.

We calculate the summer warmth index (SWI) and snow-free season length using WLDAS-based estimates of daily air temperature and snow cover fraction. We consider the snow-free season length (interchangeably, the ‘growing season’) to be the number of days each year that a grid cell had less than 10% snow-cover. SWI is calculated as the sum of the monthly mean growing degree days (base zero degrees Celsius) from June to September. To quantify the direction and rate of change in WLDAS-based estimates of snow-free season length and SWI, we used non-parametric Mann–Kendall tests for monotonic trends on pre-whitened time series and quantified the trend’s rate of change using the Sen’s slope (Sen 1968) as described by Zhang *et al* (2000) for each Landsat pixel.

Informed by the results of previous studies which found that changes in remotely sensed NDVI were associated primarily with changes in temperature and snow-cover we use the rate of change of SWI

(dSWI) and number of annual snow-free days (dSFD) to predict the probability of greening, browning, or demonstrating no trend. To this end, we fit a Bayesian multinomial logistic regression with a categorical distribution and SoftMax link function using equation (3). Holding the ‘no trend’ category as the reference, the model estimates the probability of a site being classified as ‘no trend,’ ‘browning,’ or ‘greening,’ given the rate of change (Sen’s slope) in the SWI (dSWI) and snow-free season length (dSFD):

$$\begin{aligned}\mathbf{trend}_i &\sim \text{Categorical}(\pi_i), \\ \pi_{ik} &= \frac{\exp(\eta_{ik})}{\sum_{j=1}^K \exp(\eta_{jk})}, \\ \eta_{ik} &= \alpha_k + \beta_{1,k}d\text{SWI}_i + \beta_{2,k}d\text{SFD}_i, \\ \alpha_k &\sim \mathcal{N}(0, 1.5), \\ (\beta_{1,k}, \beta_{2,k}) &\sim \mathcal{N}(0, 1)\end{aligned}\tag{3}$$

where $\mathbf{trend}_i$ is a vector of probabilities that site i belongs to classification $k \in K$, where $K = \{\text{browning, greening}\}$, relative to the ‘no trend’ classification. Plots of sampling algorithm diagnostics, as well as prior predictive simulations, are included in the Supplementary Information (figure S9 and S10).

2.5. Data processing and visualization

We used the LandsatTS library in R to extract Landsat data from Google Earth Engine, cross calibrate multi-satellite time series, and model NDVI_{max} (Berner *et al* 2023). WLDAS data were downloaded directly and processed in Python (Gorelick *et al* 2017, Hoyer and Hamman 2017). All other data processing, analysis, and visualization was conducted in R using a suite of libraries (Wickham *et al* 2019, R Core Team 2022, Barrett *et al* 2023, Kassambara 2023, Kay 2024a). Bayesian models were fit via Stan (Bürkner 2017, 2018, Gabry *et al* 2019, Gabry and Mahr 2024, Goodrich *et al* 2024, Kay 2024b). Maps were made using ESRI (Redlands, CA) ArcGIS Pro.

3. Results

Our analysis of plot-scale, *in situ* NDVI measurements from a four-year warming and snow addition experiment disentangled how spring snow depth and growing season temperature interact to influence alpine productivity in the Southern Rocky Mountains (Q1). Experimental warming reduced NDVI overall (mean: -0.0543 , 95% Highest Posterior Density Interval, HPDI: $[-0.0999, -0.00679]$), while snow addition had a slight positive overall effect on NDVI (mean: 0.0161 , 95% HPDI: $[-0.0286, 0.0642]$). Snow addition slightly ameliorated the overall negative effect of warming on NDVI, but the interaction between the snow addition and warming treatments was centered at zero (mean: -0.0384 , 95% HPDI: $[-0.0885, 0.00748]$), suggesting that these effects are primarily additive, rather than multiplicative (table S1). These treatment differences were clear over time and within blocks despite substantial inter-annual and spatial variation in NDVI (figure 3)

We assessed how different spectral vegetation indices and algorithms for estimating annual NDVI_{max} explained variation in field-harvested ANPP data in order to pick an appropriate vegetation index for regional trend detection. Estimates of remotely-sensed annual maximum vegetation index values were positively associated with our long-term field ANPP data (figure S5), confirming that these data provide a reliable proxy for above-ground net vegetation productivity in the Southern Rocky Mountain alpine zone (figure S5). Our comparisons of estimated annual maximum vegetation index (NDVI_{max}, EVI_{max}, EVI2_{max}, SAVI_{max}) values found that LandsatTS’s model-based estimates outperformed simpler methods that do not account for seasonality or observational frequency. Further, we found that no vegetation index consistently outperformed the others in approximate out-of-sample prediction procedures, although both NDVI_{max} and EVI_{max} were marginally more accurate (figure S6). Therefore, below we present only NDVI_{max} results for the regional Landsat analysis to align our study with other work assessing alpine and Arctic tundra greenness trends (Berner *et al* 2020, Wainwright *et al* 2020, Liu *et al* 2021, Rumpf *et al* 2022, Choler *et al* 2024).

Our analysis of 39 years (1984–2023) of Landsat NDVI data identified a mosaic of greening and browning trends across the Southern Rocky Mountain alpine tundra (Q2). Overall, we found that the annual mean NDVI_{max} of the Southern Rocky Mountain alpine has decreased 2.8% from 1984 to 2023. This seemingly small trend is the cumulation of a mixture of pronounced greening and browning regions (figure 4). Over the whole region, browning (mean: 20.17, 95% bootstrapped confidence intervals: $[19.66, 20.69]\%$) was slightly more common than greening (17.63 $[17.14, 18.12]\%$) but the

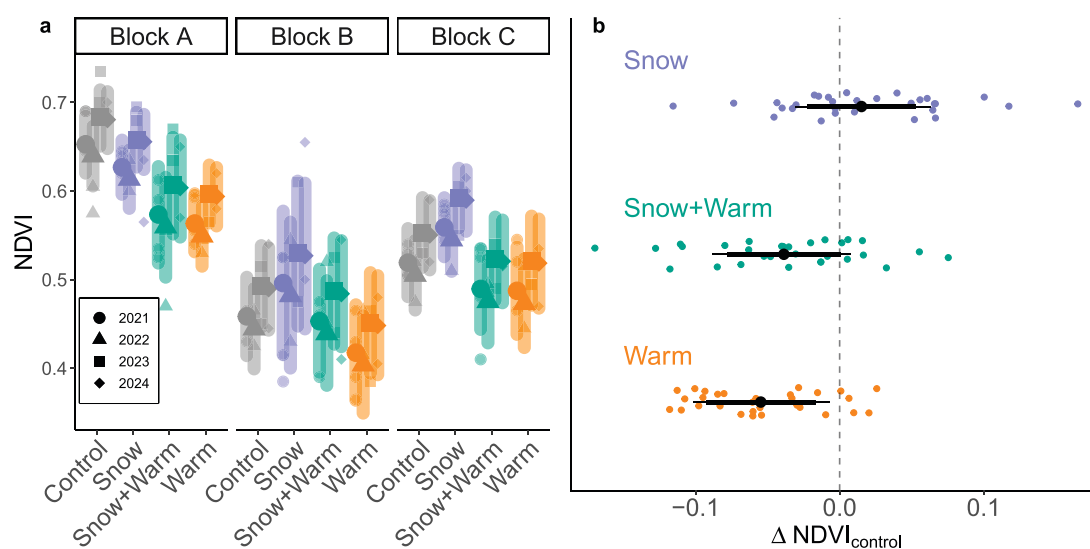


Figure 3. Experimental warming decreases productivity but extended snow cover may reduce the negative effects of warming. Model expectations of NDVI marginalized over plots within a block and treatment (a) and marginal effects of treatments (b). In (a), NDVI is shown in each block and treatment condition. Within blocks and treatments, large points and segments represent the Mean and 95% Highest Density Interval of the posterior distribution (HPDI) where each segment represents one year. In (b), points show empirical treatment effects and bars show the mean, 89% (inner), and 95% HPDI (outer) of contrasts between the marginal expectations of experimental treatments compared to controls ($\Delta \text{NDVI}_{\text{control}}$).

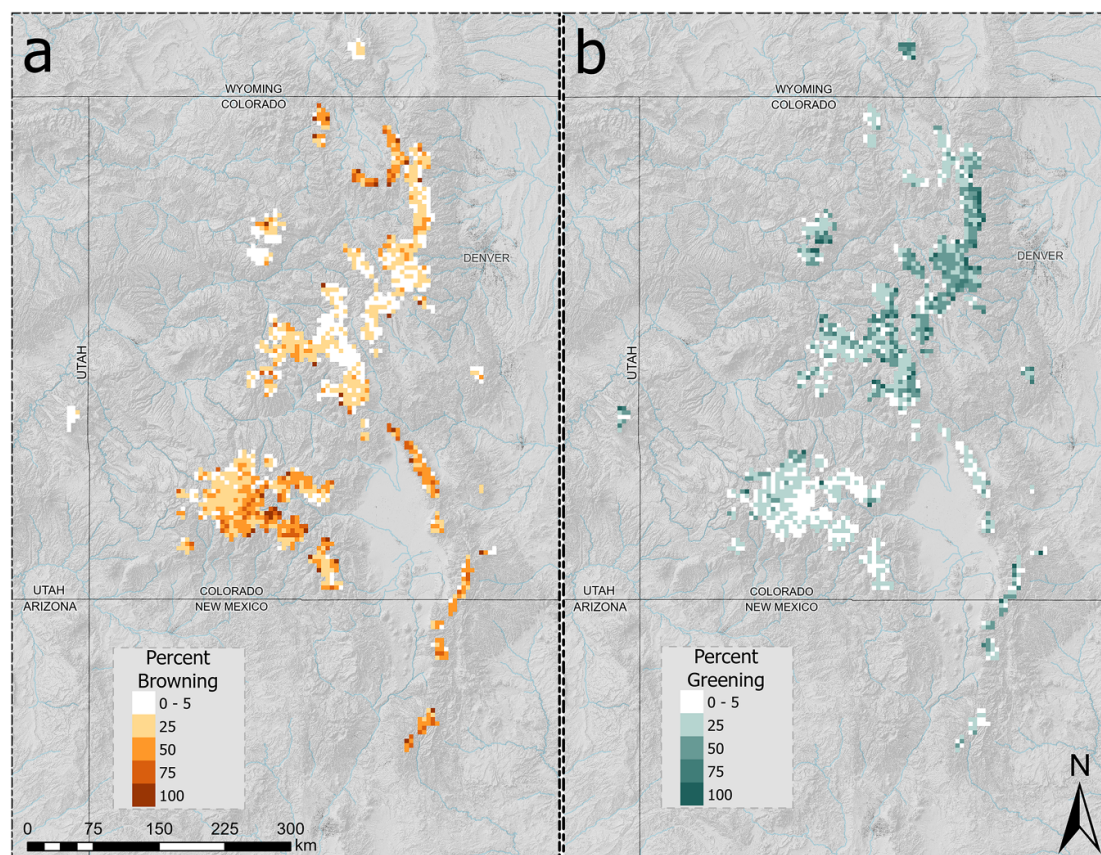
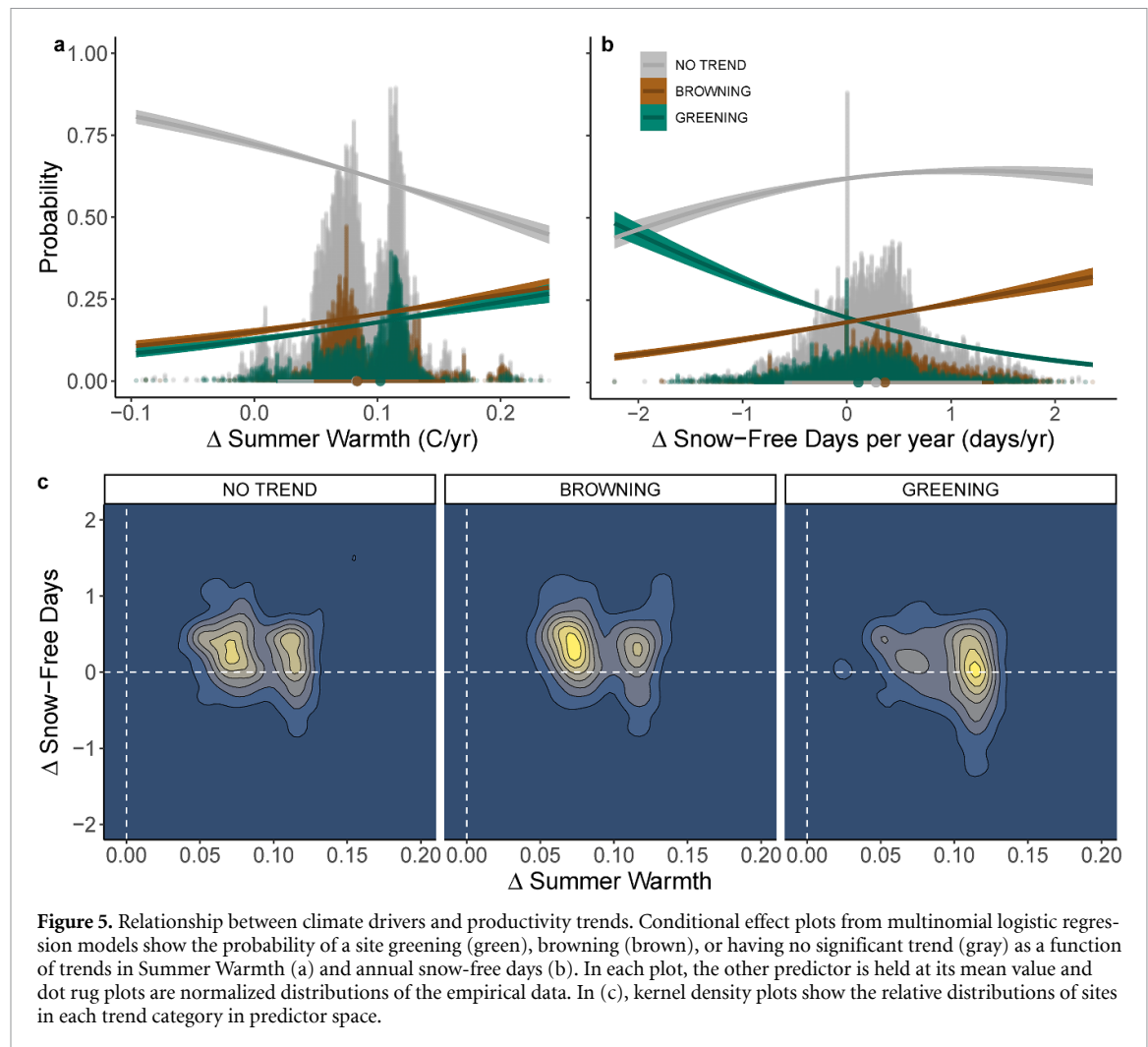


Figure 4. Distributions of relative greening and browning across the Southern Rocky Mountain alpine tundra. The proportion (%) of sampled locations that demonstrated browning (left) or greening (right) determined by pre-whitened Mann-Kendall tests for monotonic trends calculated within 5 km x 5 km² cells.



majority of the region had not changed (62.20% [61.49, 62.91]%). A visual assessment of the patterns indicates that mountain ranges further south have experienced more browning (figure 4(a)) while more northern mountain ranges have been more likely to experience greening (figure 4(b)).

Finally, we tested for covariation of greening and browning with trends in summer temperature and snow cover duration (Q3). Contrary to the expectation that warming uniformly promotes greening, we found that each additional $^{\circ}\text{C yr}^{-1}$ of summer warming increased the odds of both browning and greening by approximately 11%–12% (browning OR = 1.11, 95% CI: 1.09–1.13; greening OR = 1.12, 95% CI: 1.10–1.15; figure 5(a)), relative to no trend. Pixels experiencing greater net lengthening of the snow-free season were more likely to brown, while those with stable or shortening growing seasons were more likely to green. Moving across the interquartile range of observed snow-free season trends (-3.5 d to $+25.8$ d net change) increased the odds of browning by 17% (OR = 1.17, 95% CI: 1.13–1.21) and decreased the odds of greening by 31% (OR = 0.69, 95% CI: 0.67–0.72; figure 5(b)). The highest density of greening pixels occurred in areas with strong warming but no substantial change in the growing season length on average. Browning sites were also most common where warming was occurring (albeit more slowly on average) and the growing season was extending (figure 5(c)).

4. Discussion

Attributing trends in alpine plant productivity to specific environmental drivers requires approaches that can disentangle the relative influences of climate, soil, topography, and their interactions. Our study leveraged experimental manipulations, long-term *in situ* and satellite records, and gridded meteorological data to synthetically address these processes. Experimental data indicate that warming reduces greenness on Niwot Ridge, while snow addition has a marginal positive effect on greenness (Q1), and that these effects were primarily additive. Long-term satellite greenness records revealed an overall slight decrease

in productivity across the entire Southern Rockies, but strong spatial heterogeneity at regional scales, with some mountain ranges browning while others were greening (Q2). This heterogeneity was associated with regional climate patterns: browning tended to occur in areas that are experiencing warmer and longer growing seasons than they have had in the past, while greening occurred in areas that were warming fastest but the growing season length (snow-free duration) was stable (Q3). Together, these results help pinpoint the critical roles of temperature and growing season length in mediating alpine ecosystem responses to warming, particularly in semi-arid regions like the Southern Rocky Mountains.

Our results are consistent with the hypothesis that trends in alpine productivity in response to rising temperatures will depend on water and energy limitations, which are mediated by soil properties and topography. The absence of widespread greening in the Southern Rocky Mountains contrast sharply with the growing body of literature linking warmer and longer growing seasons to tundra greening in both the Arctic and global alpine regions, including the Pyrenees, Alps, and Greater Caucasus (Carlson *et al* 2017, Arndt *et al* 2019, Berner *et al* 2020, Rumpf *et al* 2022, Choler *et al* 2024). Moisture-contingent productivity responses have been reported in the Tibetan Plateau, where (in contrast to the NDVI trends we found here) wetter regions exhibited stronger browning while drier regions experienced greening under warming (Zhu *et al* 2023). Together, these studies emphasize the importance of moisture as a moderator of productivity responses to warming (e.g. Teng *et al* (2021)), though its specific effect varies with regional context. The perception that warming is uniformly associated with greening across alpine tundra ecosystems is likely a byproduct of the locations it has been most thoroughly measured rather than a general rule. Furthermore, even areas that are currently experiencing greening may begin to experience browning in the future (e.g. Liu *et al* (2021)).

The implications of alpine greening and browning depend on the particular biotic changes underlying the changes in NDVI. At the physiological level, patterns of greening and browning may reflect carbon allocation trade-offs under varying degrees of water limitation (Zhu *et al* 2023) and evapotranspiration responses to moisture stress (Liu *et al* 2021). At the community scale, warming tends to increase plant height and growth rates or cause shifts in community dominance, such as shrub encroachment, any of which can drive increases in remotely sensed NDVI (greening) in the Arctic tundra (Forbes *et al* 2010, Boelman *et al* 2011, Elmendorf *et al* 2012, Bjorkman 2018, Myers-Smith *et al* 2020). Though these mechanisms have been less well studied in the alpine, warming and snow-addition together have generally been shown to decrease species diversity and community productivity (Walker 2006, Wipf *et al* 2009, Farrer *et al* 2015) while snow-addition alone tends to increase productivity, likely due to reduced water limitation (Farrer *et al* 2015). Such vegetation changes can have cascading impacts on albedo and snow-trapping, with implications for surface-atmosphere energy exchanges and water availability (Pearson *et al* 2013, Loranty *et al* 2018). Alpine tundra in the dry meadows of Niwot Ridge are currently a net carbon source (Knowles *et al* 2019), so further decreases of above-ground productivity (browning) may decrease gross primary productivity, to drive additional carbon loss in these systems. Anticipating where and when these changes occur will be critical for predicting alpine responses to climate change, and how it will differ across fine-scale topographic gradients that vary in snow accumulation patterns, snowmelt timing, and community composition. For example, alpine-restricted species show greater sensitivity to snow melt timing and soil moisture dry-down than do wide-ranging generalists and may therefore serve as informative indicator species for community-wide changes (Winkler *et al* 2018).

Recent observations have documented substantial declines in summer soil moisture in the Southern Rockies, and simulations of future climates predict continued decreases throughout this century (Dettinger *et al* 2015, Bolinger *et al* 2023). Furthermore, regional ecosystem model simulations calibrated to this region predict a decoupling of the timing of vegetation water demand from supply by the end of the century (Jay *et al* 2023). Together, these studies indicate substantial vulnerability of the alpine tundra of the Southern Rockies to future climatic change. The strong association of browning trends with increasing growing-season length found here (figure 5(b))—together with other studies in the Southern Rockies (Winkler *et al* 2016, Jabis *et al* 2020, Wainwright *et al* 2020, Elias *et al* 2021)—suggest that reductions in soil moisture due to longer snow-free seasons may already be limiting productivity across much of the region. The consequences of greening and browning in the alpine extend beyond the alpine zone itself. In the Upper Rio Grande Basin, maximum snow water equivalent has advanced by approximately three weeks since 1980, impacting downstream water, and riparian vegetation along with alpine ecosystems (Barnhart *et al* 2016, Elias *et al* 2021). With roughly 60 million people in the western U.S. depending on snowmelt-derived water supply, understanding the coupled vegetation-hydrology dynamics of the Southern Rockies carries stakes well beyond ecosystem ecology (Bales *et al* 2006).

Several caveats in our analysis highlight important avenues for future research. First, the correspondence between field biomass clips and Landsat-derived NDVI data was imperfect. Challenges inherent to inter-comparisons between field-harvested and remote-sensing data including a scale mismatch between

clip harvests and the 30 m pixel resolution of Landsat imagery, as well as compounding complications due to cloud cover, atmospheric contamination, and sparse temporal coverage (Myers-Smith *et al* 2020). Validating remotely-sensed vegetation indices with long-term field data is critical for interpretation of greening or browning trends, but often omitted due to lack of field data. Second, although our study demonstrates the importance of snow-cover in explaining greening and browning, precise inference about the mechanisms underlying regional greening dynamics are constrained by the absence of high-quality soil moisture data at appropriate scales. The gridded soil moisture products currently available lack the spatial resolution and validation that is required to rigorously test this mechanism in complex alpine terrain. New satellite technology, including the recently-launched NASA-ISRO Synthetic Aperture Radar (NISAR) may facilitate deeper investigation of the role of soil moisture in constraining alpine greening in future studies. Similarly, finer-resolution remote sensing platforms that have come online in the last decade (e.g. Sentinel-2 or commercial products such as those from Planet Labs) offer improved spatial grain and more frequent revisit times that should better capture the fine-scale topographic heterogeneity characteristic of alpine landscapes.

Observations and manipulative experiments represent two complementary approaches to understanding how alpine tundra ecosystems respond to environmental change, but a third approach, process-based modeling, is an increasingly active frontier. Land surface models offer the potential to project ecosystem trajectories beyond the temporal and spatial scope of any single study, yet tuning these models to accurately represent tundra systems remains challenging. Recent efforts at Niwot Ridge illustrate both the promise and the difficulty of this work: point-scale Community Land Model simulations with site-specific functional trait and phenology observations to represent alpine tundra growth strategies (Jay *et al* 2026) found that plant functional trait uncertainty can outweigh climate scenario uncertainty in projections of future alpine ecosystem productivity. At the Arctic scale, integration of characteristic tundra vegetation into the E3SM Land Model (Murphy *et al* 2025) showed that refined plant functional types improved landscape-level carbon fluxes, though carbon allocation parameters require further development. An exciting next phase will be to constrain these models with the vegetation composition and productivity patterns documented by remote sensing and *in situ* studies—and ultimately to use better-calibrated models to conduct *in silico* experiments that are difficult or impossible to replicate in the field.

5. Conclusions

Our study provides the first comprehensive analysis of regional alpine productivity trends in the Southern Rocky Mountains and their relationships with key climatic drivers. We detected strong divergent trends in greening and browning within the region that reflect trends in productivity, with northern ranges experiencing greening and southern regions experiencing browning between 1984 and 2023. Analysis of regional climate trends, together with a climate change manipulation experiment, suggest that snow-free season length and soil moisture may constrain potential alpine productivity gains from warming in this region. Future studies that integrate observational, remote sensing, and modeling approaches could help disentangle the mechanisms driving these divergent greening trends in other regions. In particular, expanding tundra greening research to other under-studied alpine regions will be valuable for assessing the generality of greening and browning dynamics beyond the relatively well-studied systems of Europe and Asia.

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Data availability statement

The data that support the findings of this study are openly available at the following URLs/DOIs: <https://doi.org/10.6073/pasta/16a21e3a638169d4bfe7bd2710fff201> (Suding *et al* 2024), <https://doi.org/10.6073/pasta/713e56a7af8b9593d00c5c32d6203ff2> (Walker *et al* 2025), <https://doi.org/10.5067/ABBHPUIGJH5M> (Erlingis *et al* 2024).

The code to reproduce the analyses and figures is available at <https://doi.org/10.5281/zenodo.20277502>.

Supplementary data 1 available at <https://doi.org/10.1088/2752-664X/ae6d04/data1>.

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